

Question				Marking details		Marks available													
						AO1	AO2	AO3	Total	Maths	Prac								
5	(a)	(i)		crop growth on fields increases				2	2										
				fertilisers run into waterways								✓							
				plant growth in rivers and lakes increases								✓							
				aquatic animals do not have enough oxygen								✓							
				farmers' profits increase															
award (2) for all 3 correct award (1) for any 2 correct																			
		(ii)		award (1) for statement that in general the larger the population, the more ammonia is used e.g. Asia has the largest population and the largest ammonia use / the largest population uses the most ammonia 'demand for food' is equivalent to population award (1) for recognising Africa as the exception e.g. Africa has the second largest population but the fourth largest ammonia use				2	2	2									
	(b)	(i)		ammonium sulfate do not accept ammonia sulfate			1		1										
		(ii)		<table><tr><td>plants use nitrogen to make sugar</td><td></td></tr><tr><td>plants use nitrogen to make water</td><td></td></tr><tr><td>plants use nitrogen to make oxygen</td><td></td></tr><tr><td>plants use nitrogen to make protein</td><td>✓</td></tr></table>		plants use nitrogen to make sugar		plants use nitrogen to make water		plants use nitrogen to make oxygen		plants use nitrogen to make protein	✓	1			1		
plants use nitrogen to make sugar																			
plants use nitrogen to make water																			
plants use nitrogen to make oxygen																			
plants use nitrogen to make protein	✓																		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		300 (2) if answer incorrect award (1) for $\frac{75}{15} / 5 / 0.3$		2		2	2	
	(c)	(i)		$\text{N}_2 + \boxed{3} \text{H}_2 \rightleftharpoons \boxed{2} \text{NH}_3$		1		1		
		(ii)		award (1) for any of following - the reaction is reversible - the reaction goes in both directions - the reaction can go forwards or backwards do not accept equal both ways	1			1		
		(iii)	I	iron / Fe	1			1		
			II	award (1) for any of following - increases rate of reaction - makes the reaction go faster / it go faster - makes the ammonia more quickly - makes the product more quickly - process takes less time accept 'lowers the activation energy'	1			1		
				Question 5 total	4	4	4	12	4	0